

LEO ROULEAU

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EDUCATION

Polytechnique Montreal, *Bachelor's Degree in Software Engineering* 2025 - Present

- **Expected graduation:** May 2029, **Dean's List**

College André-Grasset, *College Diploma (D.E.C) in Computer Science and Mathematics* 2023 - 2025

EXPERIENCE

Google Summer of Code, Haiku OS, *Open-Source Contributor* May 2026 - Aug 2026

- Transformed the Devices preferences application into an interactive hardware manager using C++ and the native Haiku API framework.
- Developed a hardware compatibility reporting feature using C++ and a Flask REST API, enabling users to submit system profiles and driver configurations to identify problematic devices.
- Interfaced with kernel and device drivers to extract and display detailed device types, loaded driver status, and specific /dev entry mappings.
- Implemented comprehensive USB tree mapping, allowing inspection of interface descriptors and HID report formats.
- Added functionality allowing users to safely block or disable drivers for specific devices directly from the interface.

DRA Global, *Electrical Engineering Intern* May 2026 - Aug 2026

- Automated the generation of complex electrical and instrumentation drawings and single-line diagrams utilizing EPLAN Electric P8 and Excel scripting, significantly reducing manual labor and minimizing errors.
- Developed custom VBA scripts to dynamically parse engineering data and automatically render schematic layouts.
- Collaborated with Senior and Principal Electrical engineers to perform technical drawing reviews, ensuring adherence to company standards and engineering quality benchmarks.
- Presented the custom automation tools and resulting workflows to senior management and international engineering teams across the North American, South American and South African offices.

Service Télématicque Étudiant de Polytechnique (STEP), *Treasurer (Previously: VP of Cybersecurity)* Sep 2025 - Present

- Developed automated virtual machine templates using Packer and Ansible on Proxmox VE, reducing manual configuration effort and ensuring consistent environment provisioning.

EURO CAR WASH LTD, *Front-end Developer Intern* Jun 2025 - Aug 2025

- Developed and styled reusable React components to match the company's updated Figma design specifications.
- Ensured UI consistency by fixing layout issues and correcting design mismatches in Typescript.
- Collaborated in code reviews, addressing feedback and resolving errors in pull requests with my mentor.

PROJECTS & CERTIFICATIONS

C Bytecode Virtual Machine

- Engineered a stack-based bytecode VM in C, featuring a single-pass compiler and Pratt expression parser.
- Implemented a mark-and-sweep garbage collector and dynamic hash tables to handle memory management and object allocation safely.

Tree-Walk Interpreter

- Developed a complete tree-walk interpreter encompassing a custom Lexer, a recursive descent parser, and an abstract syntax tree evaluator in Java.
- Supported complex language features such as static scoping, control flow, functions, classes, and inheritance.

Certifications

- [Full-Stack Web Development Certification](#) (Udemy).

TECHNICAL SKILLS

Programming Languages: C/C++, Java, Python, JavaScript, TypeScript, VBA, Emacs Lisp

Tools & Environment: Linux, Git, Docker, Proxmox, Packer, Jira, QEMU, EPLAN Electric P8, Figma